All Network IPV4

Network	Subnet	Start	End	Related To
10.11.12.0/29	255.255.255.248	10.11.12.1	10.11.12.6	Cairo Vlan 10
10.11.12.8/29	255.255.255.248	10.11.12.9	10.11.12.14	Cairo Vlan 20
10.11.12.16/29	255.255.255.248	10.11.12.17	10.11.12.22	Cairo Vlan 30
10.11.12.24/29	255.255.255.248	10.11.12.25	10.11.12.30	Alex Vlan 40
10.11.12.32/29	255.255.255.248	10.11.12.33	10.11.12.38	Alex Vlan 50
10.11.12.40/29	255.255.255.248	10.11.12.41	10.11.12.46	Alex Vlan 60
10.11.12.48/29	255.255.255.248	10.11.12.49	10.11.12.54	Banha Vlan 70
10.11.12.56/29	255.255.255.248	10.11.12.57	10.11.12.62	Banha Vlan 80
10.11.12.64/29	255.255.255.248	10.11.12.65	10.11.12.70	Switch Cairo - Manage
10.11.12.72/29	255.255.255.248	10.11.12.73	10.11.12.78	Switch Alex- Manage
10.11.12.80/29	255.255.255.248	10.11.12.81	10.11.12.86	Switch Banha - Mange
10.11.12.88/29	255.255.255.248	10.11.12.89	10.11.12.89	Cairo Wireless Network 200
10.11.12.96/30	255.255.255.252	10.11.12.97	10.11.12.98	R- Cairo To R-Center
10.11.12.100/30	255.255.255.252	10.11.12.101	10.11.12.102	R- Alex To – R-Center
10.11.12.104/30	255.255.255.252	10.11.12.105	10.11.12.106	R- Banha To – R-Center
10.11.12.108/30	255.255.255.252	10.11.12.109	10.11.12.110	R-Center To – R-DMZ
10.11.12.116/30	255.255.255.252	10.11.12.117	10.11.12.118	R-Aswan – R-DMZ
10.11.12.120/30	255.255.255.252	10.11.12.121	10.11.12.122	R-Village – To R-DMZ
10.11.12.124/30	255.255.255.252	10.11.12.125	10.11.12.126	R-Mans To R-DMZ
10.11.12.128/29	255.255.255.248	10.11.12.129	10.11.12.134	Aswan Vlan 10
10.11.12.136/29	255.255.255.248	10.11.12.137	10.11.12.142	Aswan Vlan 20
10.11.12.144/29	255.255.255.248	10.11.12.145	10.11.12.150	Village Vlan 30
10.11.12.152/29	255.255.255.248	10.11.12.153	10.11.12.158	Village Vlan 40
10.11.12.160/29	255.255.255.248	10.11.12.161	10.11.12.166	Mans Vlan 50
10.11.12.168/29	255.255.255.248	10.11.12.169	10.11.12.174	Mans - Vlan 60
10.11.12.176/29	255.255.255.248	10.11.12.177	10.11.12.182	Aswan-Switch-Manage= 70
10.11.12.184/29	255.255.255.248	10.11.12.185	10.11.12.190	Village-Switch-Manage = 80
10.11.12.192/29	255.255.255.248	10.11.12.193	10.11.12.198	MANS – Switch -Manage = 90
10.11.12.200/29	255.255.255.248	10.11.12.201	10.11.12.206	Fire Wall – Inside Zone
10.11.12.208/29	255.255.255.248	10.11.12.209	10.11.12.214	Fire Wall – DMZ Zone
10.11.12.216/29	255.255.255.248	10.11.12.217	10.11.12.222	R-DMZ To - Firewall
10.11.12.224/29	255.255.255.248	10.11.12.225	10.11.12.230	
10.11.12.232/29	255.255.255.248	10.11.12.233	10.11.12.238	
10.11.12.240/29	255.255.255.248	10.11.12.241	10.11.12.246	
10.11.12.248/29	255.255.255.248	10.11.12.249	10.11.12.254	
10.11.12.112/30	255.255.255.252	10.11.12.113	10.11.12.114	

Cairo Branch

Vlans:

- VLAN 10 (HR) is used for the Human Resources department network
- VLAN 20 (IT) is assigned to the Information Technology department
- VLAN 30 (OPS) is used for the Operations department
- VLAN 90 (Cairo-Manage) is dedicated to network and switch management
- VLAN 200 (WIFI-Access) is used for wireless and guest users.
- Tacacs Server on Vlan 10 in switch S-Right Cairo with Ip address = 10.11.12.2

Vlan Name	Network
HR =10	10.11.12.0/29
IT = 20	10.11.12.8/29
OPS = 30	10.11.12.16/29
Cairo-Mange = 90	10.11.12.64/29
WIFI-Access = 200	10.11.12.88/29
Cairo-Branch = 100	Native

Switch:

- Switches provide Layer 2 connectivity for end devices
- User name is =noor, password =noor, root user
- Switches are named S-Cairo-Center, S-Left-Cairo, and S-Right-Cairo
- Implement Layer 2 security, 802 Authentication for ports, all port that un used in Black Hole Vlan
- Manage all Vlans By S-Cairo-Center it's server for VTP.

Switch Name	IP Address
S-Cairo-Center	10.11.12.66
S-Left-Cairo	10.11.12.67
S-Right-Cairo	10.11.12.68

Router-Cairo:

- DHCP service is enabled on the router
- DHCP automatically assigns network configuration to hosts
- Each VLAN has its own DHCP scope.
- Router connect to Tacacs Server (10.11.12.2) for Authentication.
- If Tacacs down Router connect to Radius Server (10.11.12.42) for Authentication.
- If Tacacs and Radius server down Login with local account

Interface	IP	Related To
G0/0/0.10	10.11.12.1	Vlan 10
G0/0/0.20	10.11.12.9	Vlan 20
G0/0/0.30	10.11.12.7	Vlan 30
G0/0/0.90	10.11.12.65	Vlan 90
G0/0/0.200	10.11.12.89	Vlan 200
S0/1/1	10.11.12.98	Outside To Main Router

Alex Branch

Vlans:

- VLAN 40 (FIN) is used for the Human Resources department network.
- VLAN 50 (Sales) is assigned to the Information Technology department.
- VLAN 60 (RED) is used for the Operations department.
- VLAN 100 (Alex-Manage) is dedicated to network and switch management.
- Radius Server on Vlan 60 with static IP 10.11.12.42

Vlan Name	Network
FIN =40	10.11.12.24/29
Sales = 50	10.11.12.32/29
RED = 60	10.11.12.40/29
Alex-Mange = 100	10.11.12.72/29
Alex-Branch = 110	Native

Switch:

- Switches provide Layer 2 connectivity for end devices.
- Switches are named S-Alex-Center, S-Alex-Cairo, and S-Alex-Cairo.
- Implement Layer 2 security, 802 Authentication for ports, all port that un used in Black Hole Vlan.
- Manage all Vlans By S-Alex-Center it's server for VTP.
- User name is =noor, password =noor,

Switch Name	IP Address
S-Alex-Center	10.11.12.74
S-Left-Alex	10.11.12.75
S-Right-Alex	10.11.12.76

Router-Cairo:

- DHCP service is enabled on the router.
- DHCP automatically assigns network configuration to hosts
- Each VLAN has its own DHCP scope.
- Router connect to Radius Server (10.11.12.42) for Authentication.
- If Radius down Router connect to Tacacs Server (10.11.12.2) for Authentication.
- If Tacacs and Radius server down Login with local account.
- User name is =noor, password =noor local user account.

Interface	IP	Related To
G0/0/0.40	10.11.12.25	Vlan 40
G0/0/0.50	10.11.12.33	Vlan 50
G0/0/0.60	10.11.12.41	Vlan 60
G0/0/0.100	10.11.12.73	Vlan 90
S0/1/0	10.11.12.102	Outside To Main Router

Banha Branch

Vlans:

- VLAN 70 (SEC) is used for the Human Resources department network
- VLAN 80 (BLUE) is assigned to the Information Technology department
- VLAN 110 (Banha-Manage) is dedicated to network and switch management
- NTP Server 10.11.12.50 – Syslog Server = 10.11.12.51 both on Vlan 70

Vlan Name	Network
SEC = 70	10.11.12.48/29
BLUE = 80	10.11.12.56/29
Banha-Mange = 110	10.11.12.80/29
B-Banha	Native

Switch:

- Switches provide Layer 2 connectivity for end devices
- Three switches are deployed in the Cairo site
- Switches are named S-Banha-Center, S-Left- Banha, and S-Right- Banha
- A dedicated management VLAN is used for switch administration
- Manage all Vlans By S- Banha -Center it's server for VTP
- User name is =noor, password =noor,

Switch Name	IP Address
S- Banha-Center	10.11.12.82
S-Left- Banha	10.11.12.83
S-Right- Banha	10.11.12.84

Router- Banha:

- DHCP service is enabled on the router
- DHCP automatically assigns network configuration to hosts
- Each VLAN has its own DHCP scope
- DHCP simplifies network administration
- Design vpn connection for network 10.11.12.56
- User name is =noor, password =noor, enable password=noor root users
- Level 5 user username=Mohamed, password=noor, enable 5 password= noor
- Root View enable view password=noor, enable view support = noor

Interface	IP	Related To
G0/0/0.70	10.11.12.49	Vlan 70
G0/0/0.80	10.11.12.57	Vlan 80
G0/0/0.110	10.11.12.81	Vlan 110
S0/1/0	10.11.12.106	Outside To Main Router

Router Center for City Network

Router Center:

- Design zone policy inside, outside.
- All interfaces s0/1/1, s0/2/0, s0/2/1 all inside Zone
- The interface s0/1/0 in Outside Zone.
- The Traffic from inside to outside http, ftp, https, icmp with inspect action, vpn pass.
- The Traffic form outside to inside vpn pass.
- Create ospf authentications for serial interfaces.
- User name is =noor, password =noor, enable password=noor

Interface	IP Address
S0/1/0 – DMZ Router	10.11.12.109
S0/1/1 – Cairo Router	10.11.12.97
S0/2/0 – R-Alex	10.11.12.101
S0/2/1 – R-Banha	10.11.12.105

Router DMZ:

- User name is =noor, password =noor, enable password=noor

Interface	IP Address
S0/1/0 – Router Center	10.11.12.110
S0/1/1 – R - Aswan	10.11.12.117
S0/2/0 – R-Village	10.11.12.121
S0/2/1 – R-Mans	10.11.12.126
G0/0/0 To Fire Wall	10.11.12.217

Aswan Branch

Vlans:

- VLAN 10 (Soc) is used for the Human Resources department network
- VLAN 20 (Admin) is assigned to the Information Technology department
- VLAN 70 (Awan-Manage) is dedicated to network and switch management

Vlan Name	Network ipv4	Network ipv6
SOC = 10	10.11.12.128/29	2000:12:34:1::/64
ADMIN = 20	10.11.12.136/29	2000:12:34:2::/64
Aswan-Mange = 70	10.11.12.176/29	2000:12:34:7::/64
Aswan-Native = 110	Native	

Switch:

- Switches provide Layer 2 connectivity for end devices
- Three switches are deployed in the Cairo site
- Switches are named S-Aswan -Center, S-Left- Aswan, and S-Right- Aswan
- A dedicated management VLAN is used for switch administration
- Manage all Vlans By S- Aswan-Center it's server for VTP
- User name is =noor, password =noor, root user

Switch Name	IPv4	Ipv6
S- Aswan -Center	10.11.12.178	2000:12:34:7::2
S-Left- Aswan	10.11.12. 179	2000:12:34:7::3
S-Right- Aswan	10.11.12. 180	2000:12:34:7::4

Router- Aswan:

- DHCP service is enabled on the router
- DHCP automatically assigns network configuration to hosts
- Each VLAN has its own DHCP scope
- DHCP simplifies network administration
- User name is =noor, password =noor, root user
- Design vpn connection for network 10.11.12.128, 10.11.12.136

Interface	IPv4	IPv6	Related To
G0/0/0.10	10.11.12.129	2000:12:34:1::1	Vlan 10
G0/0/0.20	10.11.12.137	2000:12:34:2::1	Vlan 20
G0/0/0.70	10.11.12.177	2000:12:34:7::1	Vlan 70
S0/1/0	10.11.12.106		Outside To DMZ Router

Village Branch

Vlans:

- VLAN 30 (IR) is used for the Human Resources department network
- VLAN 40(THREAT) is assigned to the Information Technology department
- VLAN 80(Village-Manage) is dedicated to network and switch management

Vlan Name	IPV4	IPV6
IR = 30	10.11.12.142/29	2000:12:34:3::/64
THREAT = 40	10.11.12.152/29	2000:12:34:4::/64
Village-Mange = 80	10.11.12.184/29	2000:12:34:8::/64
Village-Native = 120	Native	

Switch:

- Switches provide Layer 2 connectivity for end devices
- Three switches are deployed in the Cairo site
- Switches are named S- village -Center, S-Left- village, and S-Right- village
- A dedicated management VLAN is used for switch administration
- Manage all Vlans By S- village -Center it's server for VTP

Switch Name	Ipv4	Ipv6
S- village-Center	10.11.12.186	2000:12:34:8::2
S-Left- village	10.11.12.187	2000:12:34:8::3
S-Right- village	10.11.12.188	2000:12:34:8::4

Router- Village:

- DHCP service is enabled on the router
- DHCP automatically assigns network configuration to hosts
- Each VLAN has its own DHCP scope
- DHCP simplifies network administration

Interface	IPV4	IPV6	Related To
G0/0/0.30	10.11.12.145	2000:12:34:3::1	Vlan 30
G0/0/0.40	10.11.12.153	2000:12:34:4::1	Vlan 40
G0/0/0.80	10.11.12.185	2000:12:34:8::1	Vlan 80
S0/1/0	10.11.12.122		Outside To Main Router

Mans Branch

Vlans:

- VLAN 50 (HELPDESK) is used for the Human Resources department network
- VLAN 60 (SUPPORT) is assigned to the Information Technology department
- VLAN 30 (Mans -Manage) is dedicated to network and switch management

Vlan Name	IPV4	IPV6
HELPDESK = 50	10.11.12.160/29	2000:12:34:5::/64
SUPPORT = 60	10.11.12.168/29	2000:12:34:6::/64
Mans-Mange = 90	10.11.12.192/29	2000:12:34:9::/64
B- Mans	Native	

Switch:

- Switches provide Layer 2 connectivity for end devices
- Three switches are deployed in the Cairo site
- Switches are named S- Mans -Center, S-Left- Mans, and S-Right- Mans
- A dedicated management VLAN is used for switch administration
- Manage all Vlans By S- Mans -Center it's server for VTP
- User name is =noor, password =noor, root user

Switch Name	IPV4	IPV6
S- Mans-Center	10.11.12.194	2000:12:34:9::2
S-Left- Mans	10.11.12.195	2000:12:34:9::3
S-Right- Mans	10.11.12.196	2000:12:34:9::4

Router- Mans:

- DHCP service is enabled on the router
- DHCP automatically assigns network configuration to hosts
- Each VLAN has its own DHCP scope
- DHCP simplifies network administration
- User name is =noor, password =noor, root user

Interface	IPV4	IPV6	Related To
G0/0/0.50	10.11.12.49	2000:12:34:5::1	Vlan 50
G0/0/0.60	10.11.12.57	2000:12:34:6::1	Vlan 60
G0/0/0.90	10.11.12.193	2000:12:34:9::1	Vlan 90
S0/1/1	10.11.12.125		Outside To Main Router

Fire Wall Zone

Inside Network:

- Inside Network is 10.11.12.200
- Security level 100

DMZ Network:

- DMZ Network is 10.11.12.208
- Security Level is 50
- DMZ Server with Static IP 10.11.12.211

Rules:

- access-list DMZ-list extended permit icmp any any echo-reply.
- access-list DMZ-list extended permit tcp any eq www any.
- access-list DMZ-list extended permit tcp any eq 443 any.
- access-list OUTSIDE_ACL extended permit tcp any host 10.11.12.211 eq www.
- access-list OUTSIDE_ACL extended permit tcp any host 10.11.12.211 eq 443.
- access-list OUTSIDE_ACL extended permit tcp any host 10.11.12.211 eq smtp
- Nating From outside To DMZ With IP 10.11.12.219.